

On graph based pseudo quadratic multivariate maps of prescribed degree as instruments of key establishment.

Vasyl Ustimenko¹, Tymoteusz Chojecki²

¹ Royal y Holloway University of London and Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, e-mail: Vasyl.Ustymenko@rhul.ac.uk

² Institute of Computer Science and Mathematics, UMCS pl. Marii Curie-Skłodowskiej 1, 20-031, Lublin, Poland, Email: tymoteusz.chojecki@umcs.pl

Abstract. Let us assume that one of two trusted parties (administrator) manages the information system (IS) and another one (user) is going to use the resources of this IS during the certain time interval. So they need establish secure user's access password to the IS resources of this system via selected authenticated key exchange protocol. So they need to communicate via insecure communication channel and secretly construct a cryptographically strong session key that can serve for the establishment of secure passwords in the form of tuples in certain alphabet during the certain time interval. Nowadays selected protocol has to be postquantum secure. We propose the implementation of this scheme in terms of Symbolic Computations. The key exchange protocol is one of the key exchange algorithms of Noncommutative Cryptography with the platform of multivariate transformation of the affine space over selected finite commutative ring. The session key is a multivariate map on the affine space. Platforms and multivariate maps are constructed in terms of Algebraic Graph Theory.

Keywords. Key establishment algorithms, Symbolic Computations, Noncommutative Cryptography, Protocol based cryptosystems, Multivariate public keys.

1. Introduction

Multivariate Cryptography (MC) is one of the fifth directions of Post- Quantum Cryptography designed for the constructions and investigations of asymmetric cryptographic algorithms with the resistance to attacks of the adversary who uses quantum computer. The security of algorithm of MC is based on the complexity of solving the system of nonlinear equations over the finite commutative ring. Traditionally MC uses systems of equations of degree 2 or 3 defined over the finite field (see [1]-[13]). Despite the fact that the last talk on Multivariate Cryptography on Eurocrypt conferences was in 2021 (see [14] and [15]) many new results in this area were obtained (see Proceedings of PQCrypto 2021-2024 and [16]-[18]) and new cryptosystem [19] were submitted to NIST (USA).

Classical task of Multivariate Cryptography is the constructions of public keys in the form of multivariate maps of small degree over finite fields. We consider more general task of construction of nonlinear multivariate map F on affine space K^n where K is a commutative ring with unity with the trapdoor accelerator T which is a piece of information such the knowledge of T allows us to compute the reimage of F in polynomial time (see [20], [21] for some examples).

We assume that multivariate map F is given in its standard form of kind $x_i \rightarrow f_i(x_1, x_2, \dots, x_n)$, $i=1, 2, \dots, n$ where polynomials f_i are given in the form of the some of monomial terms listed in the lexicographical order. We assume that monomial term M is written in the form $a(x_1)^{t(1)}(x_2)^{t(2)} \dots (x_n)^{t(n)}$, where $t(i)$ are elements of Z_m , m is the order of multiplicative group K^* .

The construction of such forms with the corresponding trapdoor accelerator is an interesting task of Algebraic Geometry for which cases of complex numbers or algebraically closed field K are especially important.

We assume that Multivariate Cryptography in wide sense is about applications of the theory of nonlinear trapdoor accelerators to Symmetric and Asymmetric Cryptography. In the case when K is a finite commutative ring the pair (F, T) can be used as instrument of symmetric encryption. The space K^n can serve as space of ciphertexts. Usually the knowledge of T allows efficient computation of the value of F on the given tuple. Both correspondents have the knowledge on T . So they do not need to compute the standard form. Adversary can use various attacks to investigate the multivariate nature of encryption procedure for the construction of the procedure to compute the reimages of the map.

Standard forms of F with trapdoor accelerator can be considered as potential multivariate maps which can serve as instruments for encryption or conducting digital signatures.

Multivariate maps are elements of Cremona semigroup ${}^nCS(K)$ of endomorphisms of $K[x_1, x_2, \dots, x_n]$ (see [22]). Such endomorphism F can be given by its values $F(x_i) = f_i(x_1, x_2, \dots, x_n)$, $i=1, 2, \dots, n$ on generic variables x_i . We assume that polynomials f_i are given via their standard form and define degree and density of the map F .

Trapdoor accelerators can be used for the generation of subsemigroup H of ${}^nCS(K)$ with the property P of computation of the product of n elements from H in a polynomial time. Its alternative approach to combinatorial method of generators and relations. Note that ${}^nCS(K)$ itself does not possess P itself because the product of n its general representatives of degree k , $k \geq 2$ will be of degree k^n .

Subsemigroup H satisfying property P can be used as platforms of Noncommutative Cryptography ([23]-[31]) for the implementation of algebraic key exchange protocols of Postquantum Cryptography. Recent cryptanalytic results on algorithms of Noncommutative Cryptography can be found in [32]-[37].

Some methods of construction of multivariate maps with the trapdoor accelerator in terms of Algebraic Graph Theory are presented in [20] together with some cryptographical applications.

The paper is dedicated to applications of graph based constructions of trapdoor accelerators to algorithm of the establishment of secure user's access password to the resources of Information System with further options of the password changes.

We suggest the following general scheme.

Assume that Administrator A and his/her trusted user have safely protected password t for mutual authentication, This is the string from K^n .

Administrator A of the Information System (IS) possesses the map F in n -variables and its trapdoor accelerator T . He/she is going to give secure access to the resources of IS to trusted user U . So A and U executes selected protocol of Noncommutative Cryptography in terms of special subsemigroup S of the affine Cremona semigroup of all multivariate maps of K^n into itself. The output of the protocol X can be used by A and U for the mutual identification. U sends $X(t)$ to A who compares it with own computation of $X(t)$.

Administrator creates the map F of the same degree $\deg(X)$ of the affine space of K^n . Administrator sends $F+X$ to U . User restores F . Now A is able to create pseudorandom or genuinely random password $(p_1, p_2, \dots, p_n) = p$ as the condition to enter the system. Administrator solves the equation $F(x)=b$ and sends the solution $x=(d_1, d_2, \dots, d_n)=d$ to the user together with the link for entering the password. User U gets the password as $F(d_1, d_2, \dots, d_n)$.

Administrator has the option to change the password several times working with the same map F with the trapdoor accelerator. He/she is able to change F via a new session of the protocol and delivery scheme. Some modifications of this procedure are discussed in the conclusion of the paper.

The security of this scheme rests on the security of selected Postquantum Protocol on Noncommutative Cryptography. We describe Twisted Diffie-Helman protocol which use the complexity of Conjugation Power Problem of the subsemigroup of ${}^nCS(K)$ satisfying property P . The general idea of this scheme is given in [38], some other protocols and platforms can be found in [39]-[42]. Some of them use the semigroup ${}^nES(K)$ of Eulerian transformations (see [43]).

This paper is dedicated to the implementation of the scheme with constructions of graph based multivariate maps of prescribed degree and density with the trapdoor accelerators. We use the platforms generated by the transformations of densities $O(1)$ or $O(n)$. Recall that the density of multivariate map F is a maximal value of densities of $f_i = F(x_i)$, $i=1, 2, \dots, n$ which are numbers of monomial terms of these multivariate polynomials. We define the global density $gden(F)$ as $den(f_1) + den(f_2) + \dots + den(f_n)$.

2. Remarks o the structure of the paper.

Section 3 is dedicated to the method of multivariate maps in terms of linguistic graphs of type $(1, 1, n-1)$. These graphs over commutative ring K defined by the system of triangular equations are used for the constructions of multivariate maps with the trapdoor accelerators since 1998 (see [44], [45], [46], [47], [48], [20] and further references). The formal definition of more general linguistic graphs appeared

Section 4 contains the definition of linguistic graphs $D(n, K)$ of type $(1, 1, n-1)$ defined over arbitrary commutative ring K with the unity. Special representatives $D(n, K)$ (see [50]) of this class were used in the case $K=F_q$ for the first constructions of graph based stream ciphers. Later these stream ciphers were generalized for the case of arbitrary finite commutative ring. These algorithm were implemented in the case of finite fields of characteristic 2, arithmetical rings by modulo power 2 and Boolean rings (see [51] and further references). The time execution and global densities of the corresponding encryption maps were compared in the cases of such commutative rings of the same order.

Earlier ([52]) an interesting obfuscation of these stream ciphers with the use of the connectivity invariants of $D(n, F_q)$ were proposed. Some stream ciphers based on the descriptions of connected components of $D(n, K)$ were implemented recently (see [53]). The multivariate public keys of degree 3 and 2 were proposed recently in [54] and [55]. Note that the quadratic public key are defined over arbitrary commutative in terms of the connectivity invariants of $D(n, K)$.

Instead of graphs $D(n, K)$ some their homomorphic images $(A(n, K)$ and $B(n, m, K)$ [56]) or modifications considered in [57]) can be used. In this section we will use Jordan-Gauss graphs defined in [57] for the constructions of multivariate maps F of high degree, the support of size $O(1)$ and density of size $O(n)$ with the trapdoor accelerator T . Recall that monomial term M is written in the form $a(x_1)^{t(1)}(x_2)^{t(2)}\dots(x_n)^{t(n)}$, where $t(i)$ are elements of Z_m , m is the order of multiplicative group K^* . The support $sup(M)$ of M as cardinality of the set $\{i | t(i): t(i) \neq 0\}$ and $sup(F)$ is the maximal value of supports of its monomial terms.

So the computation of the value of F given in its standard form takes time $O(n^2)$. Its coincides with the global density. We can construct the multivariate public key based on the pair (F, T) of degree at least d , where d is arbitrarily selected constant.

Note that the composition $G=FL$, where L is an element of $AGL_n(K)$ in a general position will have density $O(n^2)$ and support of size $O(1)$.

So the computation of the value of the map takes time $O(n^3)$. Such maps are referred as pseudo quadratic maps. Suggested cryptosystems (F, T) are natural modifications of quadratic cryptosystems proposed earlier.

Section 5 is dedicated to implementation of algorithms from previous section. The results of computer simulations are presented there. In Section 6 we consider the protocol of Noncommutative Cryptography based on the quotients of Generalised Schubert Graphs.

These combination of the protocol with the implemented algorithms for generation of multivariate map with the trapdoor accelerator gives the realization of suggested key establishment scheme. Section 7 contains conclusive remarks.

3. Linguistic graphs of type $(1, 1, n-1)$ and multivariate maps.

Missing definitions of Incidence Structures Theory or Graph Theory reader can find in [58], [59]. Let K be a commutative ring with unity. Recall that incidence structure is a triple (P, L, I) where P is

the set of points, L is the set of lines and I is a bipartite graph with the partition sets P and L . We identify I with the corresponding binary relation on the disjoint union of P and L .

Let $P=K^n$ and $L=K^n$. We identify points with tuples of kind $(x)=(x_1, x_2, \dots, x_n)$ and lines with tuples $[y]=[y_1, y_2, \dots, y_n]$. Brackets and parenthesis are convenient to distinguished type of the vertex of the graph. If (x) and $[y]$ are incident $(x)I[y]$ if and only if the following relations hold.

$$a_2x_2-b_2y_2=f_2(x_1, y_1),$$

$$a_3x_3-b_3y_3=f_3(x_1, x_2, y_1, y_2),$$

...

$$a_nx_n-b_ny_n=f_n(x_1, x_2, \dots, x_{n-1}, y_1, y_2, \dots, y_{n-1}),$$

where a_j and b_j , $j = 2, 3, \dots, n-1$ are not zero divisors, and $f_j \in K[x_1, x_2, \dots, x_{j-1}, y_1, y_2, \dots, y_{j-1}]$, $j=2, 3, \dots, n$ are multivariate polynomials with coefficients from K (see [49], [50]).

The color $\rho(x) = \rho((x))$ (and $\rho(y) = \rho([y])$) of point (x) (line $[y]$) is defined as the projection of an element (x) (respectively $[y]$) from a free module on its

initial coordinate. As it follows from the definition of linguistic

incidence structure, for each vertex of incidence graph there exists a unique neighbour of a chosen color.

We say that a linguistic graph is of Jordan-Gauss type if the map $[(x), [y]] \rightarrow (f_2(x_1, y_1), f_3(x_1, x_2, y_1, y_2), \dots,$

$f_n(x_1, x_2, \dots, x_{n-1}, y_1, y_2, \dots, y_{n-1}))$ is a bilinear map into K^{n-1} .

Thus, all f_i are special quadratic maps. In the case of Jordan-Gauss graphs, the neighbourhood of each vertex is given by the system of linear equations written in its row-echelon form.

Let $I_{n-1}=I_{n-1}(K)$ be a linguistic graph defined over the commutative ring K . Foreach $b \in K$ and $(p) = (p_1, p_2, \dots, p_n)$, there is the unique neighbour $[l] = N_b(p)$ of the point with the color b . Similarly, for each $c \in K$ and line $l = [l_1, l_2, \dots, l_n]$ there is the unique neighbour $(p) = N_c([l])$ of the line with the color c .

On the sets P and L of points and lines of the linguistic graph, we define jump operators $J = J_b(p) = (b, p_2, p_3, \dots, p_n)$ and $J = J_b([l]) = [b, l_2, l_3, \dots, l_n]$ where $b \in K$.

Let $i(1), i(2), \dots, i(k)$ be an increasing sequence of elements from $\{2, 3, \dots, n\}$ and polynomials $g_{i(s)}(x_1, x_2, \dots, x_{i(s)}) \in K[x_1, x_2, \dots, x_{i(s)}]$, $s=1, 2, \dots, k$ such that for each pair of vertexes v and w with coordinates $v_1, v_2, \dots, v_n$ and $w_1, w_2, \dots, w_n$ from the same connected component of I the equalities $g_s(v_1, v_2, \dots, v_{i(s)}) = g_s(w_1, w_2, \dots, w_{i(s)})$ for $s=1, 2, \dots, k$. In this case we refer to g_s , $s=1, 2, \dots, k$ as family of triangular connectivity invariants of the linguistic graph $I_{n-1}(K)$ of type $i(1), i(2), \dots, i(s)$.

Examples. Natural examples of Jordan -Gauss graphs of can be obtained as induced subgraphs of geometries of Kac-Moody group G . This group G contains Borel subgroups B^+ and B^- generated by root subgroups corresponding to positive roots. Standard parabolic subgroups P_i contains B^+ . The geometry of G is defined as disjoint union of $(G:G_i)$ with the incidence relation $I : gG_i IgH_j$ if the intersection $gG_i \cap hG_j$ is not an empty set and type function $t :$

$t(gG_i)=i$. As it follows from [60] (see [20] and further references) the restriction of I onto the orbits of B^- containing P_i and P_j , $i \neq j$ is Jordan Gauss graph. If the rank n is 2 this is a linguistic graph of type $(1, 1, n-1)$. In the case of the finite field F_q , $q>2$ and finite Weyl groups (A_2, B_2, G_2) these graphs are bipartite graphs of orders $2q^2, 2q^3$ and $2q^5$ correspondently.

In case when Weil group is infinite Dihedral group this graph is an infinite the corresponding q -regular graphs, but projections of partition sets on first n coordinates defines interesting Jordan-Gauss graphs $\Gamma_n(F_q)$ with well defined projective limit.

Special modifications $D(n, q)$ of $\Gamma_n(F_q)$ in the case of Kac-Moody algebras with the Cartan matrix

$$\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

provides explicit construction of graphs of large girth of Extremal Graph Theory, they are used for the constructions of LDPC codes in

satellite communications (see [61] and further references).

If we simply consider general commutative ring K we obtain Jordan - Gauss graphs $I_n = D(n, K)$, $n \geq 2$ of type $(1, 1, n-1)$ which have triangular connectivity invariants a_s , $s=1, 2, \dots, t$, $t = [(n+2)/4]-1$ of the type $i(1), i(2), \dots, i(s)$ (see [54]). These invariants define the partition of PUL into connected components if $\text{char}(K)$ is odd, i. e. two vertexes v and w are in the same connected component if and only if $a_1(v_1, v_2, \dots, v_{i(s)}) = a_s(w_1, w_2, \dots, w_{i(s)})$ for each $s=1, 2, \dots, t$. In general case for arbitrary tuple $d=(d_1, d_2, \dots, d_t)$ from K^t the totality ${}^dD(n, K)$ of tuples x with coordinates such that the conditions $a_s(x_1, x_2, \dots, x_{i(s)}) = d_s$, $s=1, 2, \dots, t$ hold is nonempty set. Edge transitivity of $D(n, K)$ implies that set of vertexes of $D(n, K)$ is divided into classes ${}^dD(n, K)$, $d \in K^t$. The restriction of binary relation I on the set ${}^dD(n, K)$ is a bipartite graph ${}^dCD(n, K)$ with partition sets isomorphic to K^{n-t} . All graphs ${}^dCD(n, K)$ are isomorphic. So we may omit index d and write simply $CD(n, K)$.

Graphs $CD(n, q) = CD(n, F_q)$, $q > 2$ were introduced in [62]. Results on the triangular invariants for graphs $D(n, K)$ are observed in [57].

In fact graphs $CD(n, q)$ are connected if $q \neq 4$. In the case of the field F_4 this graph has exactly 4 connected components for each value of n , $n > 2$.

We present the description of graphs $D(n, K)$ and their triangular connectivity invariants in the Section 3.

Algorithm 1.

Below we introduce the algorithm of generating multivariate maps on the selected partition set of linguistic graph $I_{n-1}(K)$ of type $(1, 1, n-1)$ with the triangular connectivity invariants g_s , $s=1, 2, \dots, k$.

Assume that R is the extension of commutative ring K . Then we can associate graph ${}^RI_{n-1}(K)$ with the graph $I_{n-1}(K)$ such that partition sets of new graph are isomorphic to R^n but the incidence relation is given by the same system of equations with the coefficients from K . Assume

that g_s , $s=1, 2, \dots, k$ as family of triangular connectivity invariants of the linguistic graph $I_{n-1}(K)$ of type $i(1), i(2), \dots, i(s)$.

In particular we can take list of variables $z_1, z_2, \dots, z_n$ and take $R = K[z_1, z_2, \dots, z_n]$.

We take the special point $v_0 = (z_1, z_2, \dots, z_n)$ parameter k and the sequence of colours $f_1(z_1, g_1(z_1, z_2, \dots, z_{i(s)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)})) = h_1(z_1, z_2, \dots, z_{i(s)})$,

$f_2(z_1, g_1(z_1, z_2, \dots, z_{i(s)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)})) = h_2(z_1, z_2, \dots, z_{i(s)})$,

$\dots, f_k(z_1, g_1(z_1, z_2, \dots, z_{i(s)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)})) = h_k(z_1, z_2, \dots, z_{i(s)})$, where $f_j \in K[y_1, y_2, \dots, y_{i(j)+1}]$ and the equation $f_k(z_1, a_1, a_2, \dots, a_{i(s)}) = b$ has unique solution for each tuple of parameters $a_1, a_2, \dots, a_{i(s)}$, b with coordinates from K .

Assume that we have some bijective polynomial map h on the commutative ring K .

We refer to this requirement on h_k as *reversibility condition*. We consider the walk on vertices of ${}^RI_{n-1}(K)$ with the starting point v_0 and consecutive elements $v_1, v_2, \dots, v_k$ of colours $h_1, h_2, \dots, h_k$ with the last vertex v_k with coordinates $h_k(z_1, z_2, \dots, z_{i(s)}), u_2(z_1, z_2, \dots, z_n), u_3(z_1, z_2, \dots, z_n), \dots, u_n(z_1, z_2, \dots, z_n)$.

The vertex v_k is the point if k is even and v_k is the line if k is odd. Finally we apply operator J_g , $g = h(h_k)$ to v_k and get the vertex $v = (h(h_k(z_1, z_2, \dots, z_{i(s)}), u_2(z_1, z_2, \dots, z_n), u_3(z_1, z_2, \dots, z_n), \dots, u_n(z_1, z_2, \dots, z_n)))$.

Proposition 3. 1. *The reversibility condition implies that polynomial map $F: z_1 \rightarrow h(h_k(z_1, z_2, \dots, z_{i(s)})), z_2 \rightarrow u_2(z_1, z_2, \dots, z_n), z_3 \rightarrow u_3(z_1, z_2, \dots, z_n), \dots, z_n \rightarrow u_n(z_1, z_2, \dots, z_n)$ is a bijective transformation. The information on the graph, its triangular invariants, the sequence of colours $f_1, f_2, \dots, f_k$ is a trapdoor accelerator.*

The procedure of reimage computation is the following. Assume that the value of F on some unknown tuple $(z_1, z_2, \dots, z_n)$ is $(c_1, c_2, \dots, c_n)$. We compute $h^{-1}(c_1) = d$ and take the vertex $v_k = (d, c_2, c_3, \dots, c_n)$ and the equation $f_k(z_1, g_1(z_1, z_2, \dots, z_{i(1)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)})) = d$. We compute the values $g_j(z_1, z_2, \dots, z_{i(j)})$ as $g_j(d, c_2, c_3, \dots, c_{i(j)}) = d_j$, $j=1, 2, \dots, s$. The reversibility condition allows us

to solve the equation $f_k(z_1, d_1, d_2, \dots, d_s) = d$ for the variable z . Let $z = \alpha$. We compute the values of $f_j(\alpha, d_1, d_2, \dots, d_s) = d_j, j = 1, 2, \dots, k-1$. We form the sequence of vertices with the starting point v_k and colours $d_{k-1}, d_{k-2}, \dots, d_1, \alpha$. Last vertex of this sequence is the point $(\alpha, \alpha_1, \dots, \alpha_n) = (z_1, z_2, \dots, z_n) = F^{-1}(c_1, c_2, \dots, c_n)$.

Algorithm 2.

We take the sequence of colours $h_1, h_2, \dots, h_{k-1}$ and change h_k for the constant γ . So we take the walk as the sequence $v_0, v_1, \dots, v_{k-1}$ and compute $v'_k = N_\gamma(v_{k-1}), v_{k+1} = J_h(v'_k)$ where $h = h_k$. Recall that the first coordinate of the vertex v_{k+1} is $f_k(z_1, g_1(z_1, z_2, \dots, z_{i(s)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)})) = h_k(z_1, z_2, \dots, z_{i(s)})$, coordinates $(h_k(z_1, z_2, \dots, z_n), w_2(z_1, z_2, \dots, z_n), w_3(z_1, z_2, \dots, z_n), \dots, w_n(z_1, z_2, \dots, z_n))$.

Proposition 3.2. *The reversibility condition implies that polynomial map $H: z_1 \rightarrow h_k(z_1, z_2, \dots, z_{i(s)}), z_2 \rightarrow w_2(z_1, z_2, \dots, z_n), z_3 \rightarrow w_3(z_1, z_2, \dots, z_n), \dots, z_n \rightarrow w_n(z_1, z_2, \dots, z_n)$ is a bijective transformation. The information on the graph, its triangular invariants, sequence of colours $f_1, f_2, \dots, f_k$ and constant γ is a trapdoor accelerator.*

The procedure of reimage computation is the following. Assume that the value of H on some unknown tuple $(z_1, z_2, \dots, z_n)$ is $(c) = (c_1, c_2, \dots, c_n)$. We compute $J_\gamma(c) = (\gamma, c_2, c_3, \dots, c_n) = v_{k-1}$. Recall that the first coordinate of v_{k+1} is $f_k(z_1, g_1(z_1, z_2, \dots, z_{i(s)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_s(z_1, z_2, \dots, z_{i(s)}))$.

Noteworthy that $g_1(z_1, z_2, \dots, z_{i(s)}) = g_1(\gamma, c_2, c_3, \dots, c_{i(s)}) = d_1, g_2(z_1, z_2, \dots, z_{i(2)}) = g_2(\gamma, c_2, c_3, \dots, c_{i(s)}) = d_2, \dots, g_s(z_1, z_2, \dots, z_{i(s)}) = g_s(\gamma, c_2, c_3, \dots, c_{i(s)}) = d_s$.

We take the equation is $f_k(z_1, d_1, d_2, \dots, d_s)$. The reversibility condition allows us to solve it for z_1 . Let $z_1 = \alpha$ be the solution.

We compute $f_j(\alpha, d_1, d_2, \dots, d_s) = b_j, j = 1, 2, \dots, k-2$ and take the path of vertexes of the graph with starting vertex v_{k-1} and consecutive colours $b_{k-2}, b_{k-3}, \dots, b_1, \alpha$. The last vertex is $(z_1, z_2, \dots, z_n) = (\alpha, \alpha_2, \alpha_3, \dots, \alpha_n) = H^{-1}(c_1, c_2, \dots, c_n)$.

Examples of h_k satisfying the reversibility condition:

$(pr_1(g_1, g_2, \dots, g_s) + 1)z_1 + r_2(g_1, g_2, \dots, g_s)$ where $r_1, r_2 \in Z_q[y_1, y_2, \dots, y_s]$ in the case $K = Z_q, q = p^m$; $h(r_1(g_1, g_2, \dots, g_s))(z_1)^t + r_2(g_1, g_2, \dots, g_s)$, where $r_1, r_2 \in F_q[y_1, y_2, \dots, y_s]$, $h(y)$ has no linear terms in its decomposition in $F_q[y]$ in the case of $K = F_q, q = p^m, (t, q) = 1$; $\alpha z_1 + r(g_1, g_2, \dots, g_s)$ where $\alpha \in K^*, r \in K[y_1, y_2, \dots, y_s]$ in the case of general commutative ring with unity.

If linguistic graph coincides with $D(n, K)$ or their modifications presented in [61] then their connectivity invariants can be used for the constructions of multivariate maps with prescribed degree and density.

4. Jordan-Gauss graphs $D(n, K)$ and their modifications.

Jordan-Gauss graph $D(n, K)$ of type 1, 1, $n-1$ is defined as incidence structure I_{n-1} with the partition sets isomorphic to K^n . The point $(p) = (x_1, x_2, \dots, x_n)$ of this graph is incident with the line $[y] = [y_1, y_2, \dots, y_n]$, if the following relations between their coordinates hold:

$$\begin{aligned} x_2 - y_2 &= y_1 x_1, \\ x_3 - y_3 &= y_2 x_1, \\ x_4 - y_4 &= y_1 x_2, \\ x_i - y_i &= y_1 x_{i-2}, \\ x_{i+1} - y_{i+1} &= y_{i-1} x_1, \\ x_{i+2} - y_{i+2} &= y_i x_1, \\ x_{i+3} - y_{i+3} &= y_1 x_{i+1} \text{ where } i \geq 5. \end{aligned}$$

As it is easy to see the projective limit $D(K)$ is well defined. In fact if K is an integral domain the biregular graph $D(K)$ is the forest [50].

Recall that graphs $D(n, F_q) = D(n, q)$ are the modifications of $\Gamma_n(F_q)$ in the case of Kac-Moody algebras with the Dynkin diagram $Ext A_l$ (A_l with wave or extended Dynkin diagram of A_l). It is convenient to use positive roots of this root system as indexes of points and lines. The real roots are

$k+1\alpha_1+k\alpha_2$, $k\alpha_1+(k+1)\alpha_2$ where $k \geq 0$ α_1 , α_2 are simple roots and the imaginary roots are $k\alpha_1+k\alpha_2$ where $k \geq 1$. We identify roots with their coordinates in the lattice generated by simple roots $(k+1, k)$, $(k, k+1)$ and (k, k) . The modification uses "twins" of imaginary roots indexed as $(k, k)'$.

So graph $D(K)$ can be defined as the incidence with lines $[y]=[y_{01}, y_{11}, y_{12}, y_{21}, y_{22}, y'_{22}, \dots, y'_{ii}, y_{i+1}, y_{i+1,i}, y_{i+1,i+1}, \dots]$, points $(x)=(x_{10}, x_{11}, x_{12}, x_{21}, x_{22}, x'_{22}, \dots, x'_{ii}, x_{i+1}, x_{i+1,i}, x_{i+1,i+1}, \dots)$ and incidence relation given by equations

$$\begin{aligned} x_{ii}-y_{ii} &= x_{10} y_{i-1,i}; \\ x'_{ii}-y'_{ii} &= x_{i,i-1} y_{01}; \\ x_{i,i+1}-y_{i,i+1} &= x_{ii} y_{01}; \quad (1) \\ x_{i+1,i}-y_{i+1,i} &= x_{10} y'_{ii}. \end{aligned}$$

This four relations are defined for $i \geq 1$, $(x'_{11} = x_{11}, y'_{11} = y_{11})$.

Note that tuples (x) and $[y]$ have finite support, i. e. only finite number of their coordinates differ from zero. Coordinates x'_{ii} and y'_{ii} correspond to the root $(i, i)'$.

Further we interpret $D(n, K)$ as homomorphic images of $D(K)$ obtained via the projection of points and lines into their first n coordinates. The incidence of points and lines of $D(n, K)$ is defined by the first $n-1$ equations of the system (1).

For the description of triangular connectivity invariants of $D(n, K)$, it will be convenient for us to define $x_{-1,0} = y_{0,-1} = x_{1,0} = y_{0,1} = 0$, $x_{0,0} = y_{00} = -1$, $x'_{0,0} = y'_{0,0} = -1$, $x_{1,1} = x'_{1,1}$, $y_{1,1} = y'_{1,1}$ and to assume that our equations (1) are defined for $i \geq 0$.

Let $u = (u_0, u_{11}, u_{12}, u_{21}, \dots, u_{r,r}, u'_{r,r}, u_{r+1,r}, u_{r+1,r+1}, \dots)$, $2 \leq r \leq t$, $\alpha \in \{(1, 0), (0, 1)\}$ be a vertex of $D(k, K)$ and $a_r = a_r(u) = \sum_{i=0,r} (u_{ii} u'_{r-i, r-i} - u_{i,i+1} u_{r-i,r-i-1})$ for every r from the interval $[2, t]$, $t = \lfloor (n+2)/4 \rfloor$.

Graph $D(k, K)$ has triangular connectivity invariants $g_s = a_{s+1}(u)$, $s = 1, 2, \dots, t-1$ of type $i(1), i(2), \dots, i(s)$ where $i(1) = (2, 2)'$, $i(2) = (3, 3)'$, $\dots$, $i(t-1) = (t, t)'$.

In introduced above set Δ of not simple roots (j, j) , $(j+1, j)$, $(j, j+1)$, $(j+1, j+1)'$ where $j \geq 1$. We assume that each equation of (1) is identified by the root which appears as the index of the coordinates in the lefthand side of the equality. We consider subset Δ' of elements of $(j+1, j)$, $(j+1, j+1)'$, $j \geq 2$.

We assume that the elements of Δ are ordered accordingly to the list of coordinates of $D(K)$ in the presentation (1). Let $\Delta'(i+1, i)$ be the set of roots from Δ' which are higher than $(i+1, i)$ with respect to the defined order.

Assume that $\Delta'((i+1, i+1)')$ be the list of roots of Δ' with the order higher than $(i+1, i+1)'$. As it was proven in (i) deletion of coordinates with indexes from $\Delta'(i+1, i)$ and $\Delta'((i+1, i+1)')$ defines the homomorphism Ψ_i and Ψ'_i of graph $D(K)$. The incidence of elements of the images are defined by equations indexed by elements $\Delta - \Delta'(i+1, i)$ and $\Delta - \Delta'((i+1, i+1)')$ respectively.

The image of Ψ_i is known as graph $A(K)$. The projection of points and lines $A(K)$ on its first n coordinates defines known graphs $A(n, K)$. Graphs $A(n, K)$, $n \geq 4$ differs from $D(n, K)$.

We introduce graphs ${}^iB(K)$ and ${}^iB'(K)$ as homomorphic images of Ψ_i and Ψ'_i .

We consider projections of points and lines of these graphs onto first n coordinates together with first $n-1$ equations. The images of these homomorphisms will be denoted as ${}^iB(n, K)$ and ${}^iB'(n, K)$ for $n \geq 4i$ and $n \geq 4i+2$ respectively. These graphs were introduced in [56] in different notations. We use roots in their definitions for the description of some triangular connectivity invariants.

Proposition 4. 1. *Graphs ${}^iB(n, K)$ and ${}^iB'(n, K)$ has connectivity invariants $g_s = a_{s+1}(u)$, $s = 1, 2, \dots, i-2$ of type $j(1), j(2), \dots, j(i-2)$ where $j(1) = (2, 2)'$, $j(2) = (3, 3)'$, $\dots$, $j(i-2) = (i, i)'$.*

Let $T(i) = \{(2, 2)', (3, 3)', \dots, (i, i)'\}$ and S its proper subset, $j(S)$ is minimal number k such that $(k, k)' \in j(S)$.

We consider graphs ${}^iB_s(n, K)$ and ${}^iB'_s(n, K)$ obtained by deletion of coordinates $(r, r)'$, $(r, r) \in S$ from points and lines and replacement condition

$$x_{r+1,r} - y_{r+1,r} = x_{10} y'_{r,r} \quad \text{by} \quad x_{r+1,r} - y_{r+1,r} = x_{10} y_{r,r}$$

These graphs were introduced in [57] in different notations.

Assume that $j(S) > 2$ then ${}^iB_S(n, K)$ and ${}^iB'_S(n, K)$ have triangular connectivity invariant $g_s = a_{s+1}(u)$, $s = 1, 2, \dots, j(S)-2$. Note that partition sets of these graphs are isomorphic to K^{n-s} where s is the cardinality of S .

Let us consider the algorithms 1 and 2 in the cases of described graphs.

Assume that linguistic graph coincides with the Jordan-Gauss graphs presented in the Section 3 and multivariate maps F and H are defined via algorithms 1 and 2 respectively. Then the following propositions hold.

Proposition 4.2. *Let L be element of $AGL_n(K)$ of density $O(1)$, $k = O(1)$ and $h_1, h_2, \dots, h_k$ are functions of density $O(1)$ and degree $O(1)$. Then transformations LF has density $O(n)$ and degree $O(1)$.*

Proposition 4.3. *Let L be element of $AGL_n(K)$, $h_1, h_2, \dots, h_{k-1}$ are functions of density $O(1)$ and degree $O(1)$. Assume that h_k has density $O(n)$ and degree $O(1)$. Then transformation LH has density $O(n)$ and degree $O(1)$.*

Corollary. *Let L_1 be general element of $AGL_n(K)$. Then transformations LFL_1 and LHL_1 are pseudo quadratic. It means that for the standard forms of LFL_1 and LHL_1 the computation of their values on the given tuple costs $O(n^3)$.*

5. On implemented cases of degree 2 and 3 and other options.

The implementation of Algorithm 1 in the case of graphs $D(n, K)$ in the case of $K = F_q$, $q = 2^s$ with $h_1 = z_1 + \beta_1$, $h_2 = z_1 + \beta_2$, $h_i = h_{i-2} + \beta_i$, $i \geq 2$ and $h(x) = \alpha x^2 + \beta$, where $\beta_1 \in K$, $\beta_2 \in K$, $\beta \in K$, $\alpha \in K^*$, $\beta_i \in K^*$ for $i \geq 2$ was described in the paper [58].

The description of the implementation of Algorithm 2 in the case of $D(n, K)$, $K = F_q$, $q = 2^s$, even parameter k special colours $h_1 = \alpha_1$, $h_2 = z_{1,0} + b_1$, $h_3 = \alpha_3$, $h_4 = z_{1,0} + b_2, \dots$, $h_{k-2} = z_{1,0} + b_{k-2}$, $\alpha_{k-1} = \gamma$, $h_k = \alpha z_{1,0} + \lambda_2 a_2(z_{1,0}, z_{1,1}, \dots, z'_{2,2}) + \lambda_3 a_3(z_{1,0}, z_{1,1}, \dots, z'_{3,3}) + \dots + \lambda_t a_t(z_{1,0}, z_{1,1}, \dots, z'_{t,t})$, $t = [(n+2)/4]$, where α_i , b_i and λ_i are constants, is given in [59].

If we change $K = F_q$ for $K = Z_q$ without the change of above written requirements on h_i , $i = 1, 2, \dots$ we get the new case for the implementation. The results of corresponding computer simulations are below.

We have written a program for generating elements and for encrypting a text using above algorithm. The program is written in SAGE. We used an MacBook with a Intel Core 1,2 GHz processor, 8GB RAM, and the macOS Monterey operating system. We have implemented three **cases**: **1.** L_1 and L_2 are identities, **case 2.** L_1 and L_2 are maps of the kind $z_{1,0} \rightarrow z_{1,0} + a_2 z_{1,1} + a_3 z_{1,2} + \dots + a_t z_{1,t}$, $z_{1,1} \rightarrow z_{1,1}$, $z_{1,2} \rightarrow z_{1,2}$, $\dots$, $z_{t,t} \rightarrow z_{t,t}$, with $a_i \neq 0$, $i = 1, 2, \dots, n$ (linear time of computing for L_1 and L_2), **case 3.** $L_1 = Ax + b$, $L_2 = A_1 x + b_1$; matrices A , A_1 and vectors b , b_1 mostly have nonzero elements.

In Tables 1, 2 and 3, we describe the numbers of monomials in **case 1** for different sizes of the field. Tables 10, 11, 12 shows the generation time of the encryption.

In Tables 4, 5 and 6, we describe the numbers of monomials in **case 2** for different sizes of the field. Tables 13, 14, 15 shows the generation time of the encryption.

In Tables 7, 8 and 9, we describe the numbers of monomials in **case 3** for different sizes of the field. Tables 16, 17, 18 shows the generation time of the encryption.

Remark. After the change of the graph $D(n, K)$ on ${}^iB(n, K)$, ${}^iB_S(n, K)$, $i > 1$, $n \geq 4i$ or ${}^iB'(n, K)$, ${}^iB'_S(n, K)$ with $i > 1$, $n \geq 4n+2$ the map H is also quadratic transformation.

Table 1. Number of coefficients, ring of size 2^8 , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	141	506	1474	3794
31	141	509	1914	5676
63	141	495	1852	7270
127	141	509	1916	7333

Table 2. Number of coefficients, ring of size 2^{12} , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	141	506	1474	3794
31	141	509	1914	5770
63	141	509	1917	7418
127	141	509	1917	7421

Table 3. Number of coefficients, ring of size 2^{16} , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	141	506	1474	3794
31	141	509	1914	5770
63	141	509	1917	7418
127	141	509	1917	7421

Table 4. Number of coefficients, ring of size 2^8 , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	1021	6638	37090	175212
31	1034	6491	46899	270375
63	1075	6857	48063	367481
127	1018	6986	50212	367644

Table 5. Number of coefficients, ring of size 2^{12} , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	1076	7028	38664	172541
31	1077	7163	51443	290452

63	1078	7021	51403	391864
127	1065	7001	51073	391800

Table 6. Number of coefficients, ring of size 2^{16} , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	1078	7145	38665	178470
31	1078	7164	51758	292388
63	1078	7164	51386	391317
127	1078	7164	51764	391312

Table 7. Number of coefficients, ring of size 2^8 , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	2416	17656	136665	1068347
31	2406	17862	136663	1064700
63	2419	17834	136468	1068305
127	2434	17858	136490	1068389

Table 8. Number of coefficients, ring of size 2^{12} , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	2448	17938	137206	1072917
31	2447	17942	137198	1072934
63	2447	17947	137183	1072803
127	2447	17946	137169	1072989

Table 9. Number of coefficients, ring of size 2^{16} , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	2448	17952	137275	1073226
31	2448	17952	137278	1073261
63	2447	17951	137274	1073226
127	2448	17912	137278	1073261

Table 10. Time (ms), ring of size 2^8 , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	5	4	8	22
31	3	7	18	56

63	6	14	40	159
127	11	28	83	395

Table 11. Time (ms), ring of size 2^{12} , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	6	3	8	22
31	3	7	18	57
63	5	15	42	161
127	10	29	83	466

Table 12. Time (ms), ring of size 2^{16} , Case 1.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	5	3	7	16
31	2	5	13	41
63	5	11	31	115
127	10	20	62	298

Table 13. Time (ms), ring of size 2^8 , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	9	28	220	5602
31	7	58	688	13324
63	16	124	2008	46336
127	28	267	5034	171159

Table 14. Time (ms), ring of size 2^{12} , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	10	30	220	2981
31	8	56	661	9650
63	13	118	1890	33016
127	26	253	4608	129540

Table 15. Time (ms), ring of size 2^{16} , Case 2.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	25	172	2014	78159
31	23	192	2862	90887
63	35	257	4628	143237

127	44	379	8583	257669
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Table 16. Time (ms), ring of size 2^8 , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	25	172	2014	78159
31	23	192	2862	90887
63	35	257	4628	143237
127	44	379	8583	257669

Table 17. Time (ms), ring of size 2^{12} , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	19	136	1917	76879
31	18	167	2665	97863
63	26	236	4172	138124
127	37	365	7796	215954

Table 18. Time (ms), ring of size 2^{16} , Case 3.

	VECTOR SIZE			
PASS LENGTH	16	32	64	128
15	42	171	3065	118880
31	20	210	6001	170741
63	29	322	5208	703754
127	43	546	13970	886343

In this section we propose the generalisation of implemented scheme of Algorithm 1.

Let $I_{n-1}(K)$ be the linguistic graph of type $(1, 1, n-1)$. Assume that $g_s, s=1, 2, \dots, k$ is the family of its triangular connectivity invariants of type $i(1), i(2), \dots, i(s)$. Let $(z_1, z_2, \dots, z_n)$ be the element of pointset $(K[z_1, z_2, \dots, z_n])^n$ of $I_{n-1}(K[z_1, z_2, \dots, z_n])$.

We compute the symbolic expressions $g_s(z_1, z_2, \dots, z_{i(s)})$ for $s=1, 2, \dots, k$. We select element $z=z(y_1, y_2, \dots, y_{k+1})$ from $K[y_1, y_2, \dots, y_{k+1}]$ and compute the expression $b(z_1, z_2, \dots, z_{i(k)})=z(z_1, g_1(z_1, z_2, \dots, z_{i(1)}), g_2(z_1, z_2, \dots, z_{i(2)}), \dots, g_k(z_1, z_2, \dots, z_{i(k)}))$. We take positive integer l and consider the sequence of colours $h_1 = b(z_1, z_2, \dots, z_{i(k)}), h_2 = z_1 + \beta, \beta \in K, h_i = h_{i-2} + \beta_i, \beta_i \in K^*$.

Proposition 5. 1. Let F be the map of Algorithm 1 with the described above data in the case of one of the graphs $D(n, K)$, ${}^iB_s(n, K)$ and ${}^iB'_s(n, K)$, ${}^iB(n, K)$, ${}^iB'(n, K)$ and bijective h from $K[x]$ of degree s . Assume that degree of $b(z_1, z_2, \dots, z_{i(k)})$ is t . Then degree of F is $\max(2t+1, st)$ if l is odd and $\max(t+2, st)$ if l is even. Then degree of F is $\max(2t+1, s)$ if l is even and $\max(t+2, st)$ if l is odd.

In the case of graphs $D(n, K)$ and colours depending only from z_1 this statement is proven in [68].

In the cases of selected finite commutative rings of kind F_q or Z_q we take element L of $AGL_n(K)$ of the density $O(1)$, $L_1 \in AGL_n(K)$ and generate the pseudo quadratic standard form of $G = LFL_1$ where F satisfies the conditions of Proposition 2 with the multivariate polynomial $b(z_1, z_2, \dots, z_{i(k)})$ of density

$O(1)$. So we get multivariate public key with the pseudo quadratic map of prescribed based on the trapdoor accelerator of Proposition 1.

6. On Generalised Schubert Graphs and safe delivery of multivariate maps.

Below we present the method of construction of nonlinear representatives of the semigroup $End K[x_1, x_2, \dots, x_n]$ of multivariate maps on the affine space K^n where K is a finite commutative ring.

The incidence structure is the set V with the partition sets P (points) and L (lines) and symmetric binary relation I such that the incidence of two elements implies that one of them is a point and another one is a line. We shall identify I with the simple graph of this incidence relation which is of course a bipartite graph. The pair $x, y, x \in P, y \in L$ such that $x I y$ is called a flag of incidence structure I .

Let K be a finite commutative ring with the unity. We refer to an incidence structure with a point set $P = P_{s,m} = K^{s+m}$ and a line set $L = L_{r,m} = K^{r+m}$ as linguistic incidence structure ${}^s, {}^r I_m$ if point $x = (x_1, x_2, \dots, x_s, x_{s+1}, x_{s+2}, \dots, x_{s+m})$ is incident to line $y = [y_1, y_2, \dots, y_r, y_{r+1}, y_{r+2}, \dots, y_{r+s}]$ if and only if the following relations hold

$$\begin{aligned} a_1 x_{s+1} - b_1 y_{r+1} &= f_1(x_1, x_2, \dots, x_s, y_1, y_2, \dots, y_r), \\ a_2 x_{s+2} - b_2 y_{r+2} &= f_2(x_1, x_2, \dots, x_s, x_{s+1}, x_{s+2}, y_1, y_2, \dots, y_r, y_{r+1}), \end{aligned} \quad (1)$$

...

$$a_m x_{s+m} - b_m y_{r+m} = f_m(x_1, x_2, \dots, x_s, x_{s+1}, \dots, x_{s+m-1}, y_1, y_2, \dots, y_r, y_{r+1}, \dots, y_{r+m-1})$$

where a_j and $b_j, j=1, 2, \dots, m$ are not zero divisors, and f_j are multivariate polynomials with coefficients from K (see [49], [50]). Brackets and parenthesis allow us to distinguish points from lines.

The colour $\rho(x) = \rho((x))$ ($\rho(y) = \rho([y])$) of point (x) (line $[y]$) is defined as projection of an element (x) (respectively $[y]$) from a free module on its initial s (relatively r) coordinates. As it follows from the definition of linguistic incidence structure for each vertex of incidence graph there exists unique neighbour of a chosen colour.

We refer to $\rho((x)) = (x_1, x_2, \dots, x_s)$ for $(x) = (x_1, x_2, \dots, x_{s+m})$ and $\rho([y]) = (y_1, y_2, \dots, y_r)$ for $[y] = [y_1, y_2, \dots, y_{r+m}]$ as the colour of the point and the colour of the line respectively. For each $b \in K^r$ and $p = (p_1, p_2, \dots, p_{s+m})$ there is a unique neighbour of the point $[l] = N_b(p)$ with the colour b . Similarly for each $c \in K^s$ and line $l = [l_1, l_2, \dots, l_{r+m}]$ there is a unique neighbour of the line $(p) = N_c([l])$ with the colour c . Let $J_a(v)$ stands for the operator of change colour of vertex v (point or line) for $a = (a_1, a_2, \dots, a_t)$ where $t=s$ or $t=r$.

The triples of parameters s, r, m defines *type of linguistic graph*. We introduce the semigroup of multivariate transformation $S({}^s, {}^r I_m((K)))$ via the following steps.

S_1 . Consider the graph ${}^s, {}^r I_m(K[z_1, z_2, \dots, z_n]), n=m+s$

With the partition sets $K[z_1, z_2, \dots, z_n]^{s+m}$ (points) and $K[z_1, z_2, \dots, z_n]^{r+m}$ (lines) with the incidence given by the equations of ${}^s, {}^r I_m(K)$ with the coefficients from K .

S_2 . Consider the totality of tuples of kind ${}^k H = (g(1) = g_1(z_1, z_2, \dots, z_s), h(1) = h_1(z_1, z_2, \dots, z_s), g(2) = g_2(z_1, z_2, \dots, z_s), h(2) = h_2(z_1, z_2, \dots, z_s), g(3) = g_3(z_1, z_2, \dots, z_s), \dots, g(2k) = g_{2k}(z_1, z_2, \dots, z_s), h(2k) = h_{2k}(z_1, z_2, \dots, z_s), g(2k+1) = g_{2k+1}(z_1, z_2, \dots, z_s))$ of even length $2k+1, k=1, 2, \dots$ where $g(1) \in K[z_1, z_2, \dots, z_s]^s, h(1) \in K[z_1, z_2, \dots, z_s]^r, g(2) \in K[z_1, z_2, \dots, z_s]^r, \dots, g(2k) \in K[z_1, z_2, \dots, z_s]^r, h(2k) \in K[z_1, z_2, \dots, z_s]^s, g(2k+1) \in K[z_1, z_2, \dots, z_s]^s$ and the degree of the tuple $g(2k+1)$ is 1.

S_3 . We consider the special point $z = (z_1, z_2, \dots, z_n)$ and the sequence of vertexes

$${}^0 v = (z_1, z_2, \dots, z_n), {}^1 v = J_{g(1)}({}^0 v), {}^1 u = N_{h(1)}({}^1 v), {}^2 v = J_{g(2)}({}^1 u), {}^2 u = N_{h(2)}({}^2 v), \dots, {}^{2k} v = J_{g(2k)}({}^{2k-1} u), {}^{2k} u = N_{h(2k)}({}^{2k} v), {}^{2k+1} v = J_{g(2k+1)}({}^{2k} u).$$

Assume that ${}^{2k+1} v = (g_{2k+1}(z_1, z_2, \dots, z_s), v_2(z_1, z_2, \dots, z_n), \dots, v_n(z_1, z_2, \dots, z_n))$.

We consider the map $F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ which sends $(z_1, z_2, \dots, z_n)$ to ${}^{2k+1} v$. We define $S({}^s, {}^r I_m(K))$ as semigroup generated by transformations of kind $F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ for various tuples $(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$.

The semigroup $S({}^s, {}^r I_m((K)))$ of the transformations of K^{m+s} contains subgroup

$G({}^s, {}^r I_m(K))$ generated by maps corresponding to tuples with $g(2k+1) \in AGL_s(K)$.

Proposition 6.1. *The degree of transformation $F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ is the maximal value of $\deg h(i) + \deg g(i)$, $i=1, 2, \dots, k$.*

Generalised Double Schubert graph $DS_{s,r}(K)$ (see [20] and further references) is a bipartite graph with the points of kind $(x) = (x_1, x_2, \dots, x_s, x_{11}, x_{12}, \dots, x_{sr})$ and lines $[y] = [y_1, y_2, \dots, y_r, y_{11}, y_{12}, \dots, y_{st}]$ such that point (x) is incident to $[y]$ if and if the conditions

$$x_{ij} - y_{i,j} = x_i y_j \quad (1)$$

hold for $i=1, 2, \dots, s$ and $j=1, 2, \dots, r$.

Together with generalised Schubert graph $DS_{s,r}(K)$ with points of kind $(x) = (x_1, x_2, \dots, x_s, x_{11}, x_{12}, \dots, x_{sr})$ and lines $[y] = [y_1, y_2, \dots, y_r, y_{11}, y_{12}, \dots, y_{st}]$ we consider homomorphic image ${}^Q DS_{s,r}(K)$ where Q is selected proper subset of Cartesian product of $\{1, 2, \dots, s\} = N$ and $\{1, 2, \dots, r\} = M$. We assume that $r = O(s)$, the projection $(i, j) \rightarrow i$ maps Q onto N and the projection $(i, j) \rightarrow j$ maps Q onto M . Let $Q = \{\alpha(1), \alpha(2), \dots, \alpha(m)\}$ where $m = O(s^t)$, $1 \leq t \leq 2$. Then partition sets of ${}^Q DS_{s,r}(K)$ are affine space K^{s+m} and K^{r+m} .

We consider the map ${}^Q F = {}^Q F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ obtained in the case of linguistic graph ${}^Q DS_{s,r}(K)$. It is easy to see that $\deg {}^Q F$ is bounded by $\deg F$.

We define the bidegree of the sequence $g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ as the pair m_1, m_2 where m_1 and m_2 are the maximal values of $\deg(g(i))$, $i=1, 2, \dots, k$ and $\deg(h(i))$, $i=1, 2, \dots, k$ respectively.

We consider the semigroup $S = S({}^Q DS_{s,r}(K))$, $Q = \{k(1), k(2), \dots, k(t)\}$ of endomorphisms of the ring $K[z_1, z_2, \dots, z_s, z_{k(1)}, z_{k(2)}, \dots, z_{k(t)}]$.

Proposition 6.2. *Let ${}^m S = {}^m S({}^Q DS_{s,r}(K))$ be the subgroup generated by all ${}^Q F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1))$ such that $m_1 + m_2 \leq m$. Then the degree of S equals m .*

Let us rename variables of this multivariate ring as $y_1, y_2, \dots, y_{s+t}$ accordingly to some selected pseudorandom linear order Δ on $\{1, 2, \dots, s\}$. We take element L of $AGL_{s+t}(K)$ and consider the semigroup $H = LSL^{-1}$ of elements of kind $y_i \rightarrow h_i(y_1, y_2, \dots, y_{s+t})$, $i=1, 2, \dots, s+t$.

Let $\Psi = \Psi(\Delta, L)$ be corresponding embedding of $S({}^Q DS_{s,r}(K))$ into Cremona semigroup $End(K[y_1, y_2, \dots, y_n])$, $n=s+t$. So $\Psi({}^Q DS_{s,r}(K)) = H$.

Algorithm 3. Twisted Diffie-Hellman multivariate protocol with the hidden graph based platform.

Alice selects commutative ring K and positive integer n . She presents n as $n(1) + n(2) + n(3)$ such that $n(i) = O(n)$, $i=1, 2, 3$ where $\max(n(1), n(2)) < n(3) < n(1)n(2)$. Alice selects parameter m , $m = O(1)$. She selects order Δ on n and matrix L of density $O(1)$ from the subgroup $AGL_n(K)$ to work with elements of the subsemigroup S of

${}^m H = \Psi(\Delta, L)({}^m S({}^Q DS_{s,r}(K)))$ generated by elements of kind $\Psi({}^Q F(g(1), h(1), g(2), h(2), \dots, g(2k), h(2k), g(2k+1)))$ where the densities of the coordinates of $g(i)$, $i=1, 2, \dots, 2k+1$ and $h(i)$, $j=1, 2, \dots, 2k$ are of size $O(1)$.

It is easy to see that the density of such generator is $O(k)$. If $k = O(n^\gamma)$ then each element of S has density $O(n^\gamma)$.

Assume that Alice selects invertible element h and element g of S of densities $O(1)$. She sends g and h to her partner Bob via open channel. Next the correspondents conduct twisted Diffie Hellman protocol of Noncommutative Cryptography. Alice selects parameters $k(A)$ and $r(A)$. Alice sends $y(A) = h^{r(A)} g^{k(A)} h^{-r(A)}$ to Bob. He selects parameters $k(B)$ and $r(B)$ to compute $y(B) = h^{r(B)} g^{k(B)} h^{-r(B)}$ and send its standard form to Alice.

At the second stage of the algorithm Alice and Bob computes the collision map Y as $h^{r(A)}y(B)^{k(A)}h^{-r(A)}$ and $h^{r(B)}y(A)^{k(B)}h^{-r(B)}$ respectively.

Algorithm requires $O(1)$ multiplications in the semigroup S .

The multiplication of two elements need $O(n^{(m+1)\gamma+1})$ elementary operations. Thus the complexity of algorithm is $O(n^{(m+1)\gamma+1})$. So if $\gamma=0$ then the execution of protocol requires linear time. In case of small m , $m \geq 2$ we can select appropriate γ to get the complexity $O(n^2)$ or $O(n^3)$

Algorithm 4. Safe delivery of multivariate map of degree m .

Assume that one of the correspondents (Alice and Bob) has to deliver safely the multivariate map G from $End K[y_1, y_2, \dots, y_n]$ of degree m to his/her partner. Then they conduct above described protocol and elaborate the collision map Y . So correspondent sends $G+Y$ to his partner.

Under assumption than density of G is at least $O(n)$ these ‘‘ steganographic’’ procedure is safe.

7. Conclusion.

The problem of safe key establishment and further key management is especially important currently when solution has to be secure in sense of Postquantum Cryptography. Research on this direction can be found for instance in [63]-[67] where authors the postquantum technique different from Multivariate Cryptography.

Below discuss the option of the password establishment in term of Multivariate cryptography and its intersection with Noncommutative Cryptography. Let us use nonlinear multivariate maps of degree bounded by constant d . We assume that administrator of IS (Bob) and IS user Alice use the protocol of Noncommutative Cryptography with the multivariate platform (twisted Diffie Hellman, hidden tame homomorphism algorithms or their modification (see [38] or [20]) and the adversary is not able to break it.

For people, authentication involves setting up a username, password, and other authentication methods, such as a facial scan, fingerprint, or PIN. Passwords are hashed (not encrypted) and the hashes are saved to the database.

We assume that both parties has mutually known authentication passwords p_A (administrator) and p_B (trusted user). Tuples p_A and p_B are located in the safe data base of the system.

1 option. Authentication protocol with the multivariate platform satisfying the property P . So administrator Alice and IS user Bob have collision multivariate map H . Alice safely set the link to enter the system with the password of form $H(t)$ where t is the concatenation of p_B and p_A . Bob enters the initial password $H(t)$ and gets the access to the system. Alice can send a new access information r to Bob together with the link to access the system with the password $H(r)$. The tuple r can be of pseudorandom or genuinely random nature .

Note that Bob can make request to change the password and send r to Alice. She sets the access password as $H(r)$ and sends the link to Bob.

Alice or Bob can change the password several times. If they agree to make just single protocol adversary need to intercept quite many pairs $(r_i, H(r_i))$ and try to approximate map H . It is in fact difficult because Alice and Bob keep $H(r)$ safely. Only r_i are delivered via the open channel. In the case of quadratic map Alice can change the access password $O(n^\alpha)$, $\alpha < 2$.

The adversary can use specific features of the multivariate platform. Alice and Bob have to share some elements $g_i, i=1, 2, \dots, k$ from the platform. So adversary can try to investigate the platform generated by this elements. He/she can use specific features of the platform like a low degree and densities of elements. Sure that Alice and Bob can start new session of the protocol with other generators.

2 option. Alice can take arbitrary multivariate map F of degree at most d and sends $F+H$ to Bob. This steganographic one time pad like action is safe. So Bob Restores F . He sends $F(t)$ to Alice. She

compares the received value with the her own computation of $F(t)$ with the registered in system t . Alice takes the tuple r as above sends it to Bob and set the link with the entering condition in the form of the password $F(r)$. The complexity to form the password for both parties after conducting single protocol is $O(n^{d+1})$. To change F Alice and Bob need to start a new protocol.

3 option. Let us assume that Alice creates the bijective multivariate map F of degree d with the trapdoor accelerator T which allows her to compute the reimage in time $O(n^\alpha)$ where α is smaller than $d+1$. She sends $F+H$ to Bob. Bob sends $F(t)=c$. Alice compares $F^{-1}(c)$ with the registered t . Next Alice takes tuple r and computes $F^{-1}(r)=c$ and sends c to Bob. He restores r as $F(c)$. In this case the knowledge of T allows Alice to compute c for $O(n^\alpha)$.

4 option. Bob has (F, T) and sends $F + H$ to Alice. Bob sends $F(t)$. Alice compares it with her own independent computation of $F(t)$. She sets r and forms the link with entering condition r , computes $c=F(r)$ and sends it to Bob via open channel. So the public user can complete r as $F^{-1}(c)$ the procedure for $O(n^\alpha)$. In this case adversary has to intercept pairs (c, F^{-1}) but he has to approximate F^{-1} to get the procedure to enter the system. This is essentially harder task than the approximation of F .

We present the computer simulation just in the simplest case of quadratic transformation F defined via the equation of graph $D(n, Z_q)$ where q is the power of 2. The standard form of obtained in the case 3 of the implementation can be used as quadratic multivariate public key.

The map depends on the walk in this graph. We state the heuristic argument on the security of the key via the reference on the complexity of finding the shortest path on the graph of order v by Dijkstra algorithm. This is $v \ln(v)$ and $v=q^n$ in our case.

In fact we are getting large variety of public key because instead of $D(n, K)$ we can use other homomorphic images of infinite forests defined in [61].

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